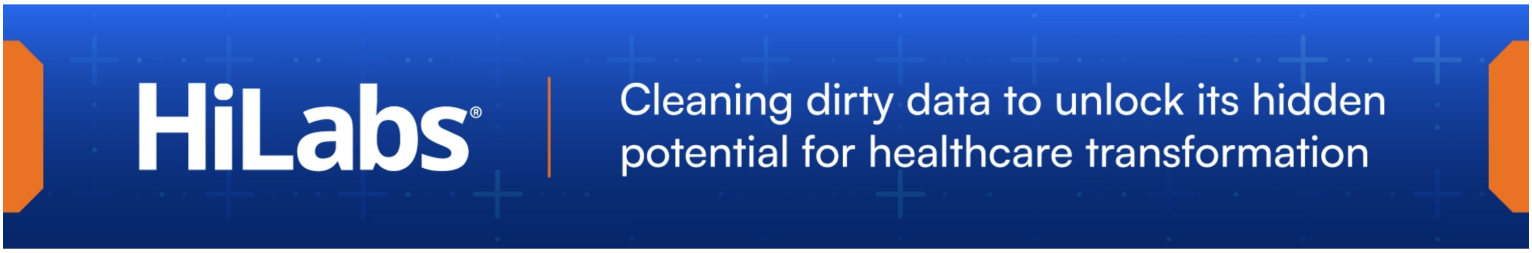
You're almost there — sign up to start building in Notion today. Sign up or login



Data Pipeline Curation for Provider Rates

1. Business Context

US health plans must publish their negotiated rates with every in-network provider as Price Transparency Data (PTD) under the Transparency in Coverage rule. Payer contracting teams use competitors' PTD to see what the market pays for a service, by provider and by location.

Getting there is a data curation problem, not a display problem:

- **Finding the right network.** A payer publishes index files covering hundreds of plans across brands and markets. Plan names rarely say which network they belong to.
- **Trusting the files.** Files have $0 rates, duplicates, stale contracts and missing codes, and there is no standard for grading them.
- **Knowing who and where.** Rates are tied to an NPI and a Tax ID (TIN), with no specialty and no address. One TIN can have many addresses and one NPI can practice at many addresses. About 80% of NPPES addresses are stale or wrong.

2. Problem Statement

Build a data pipeline that finds Centene's PPO network in its price transparency files, grades the quality of those files, and maps each provider in them to their correct practicing address, producing a provider-level rate file at the NPI, TIN, billing code and address level that a contracting team can use.

The hackathon focuses on three core problems: (1) Network Mapping, (2) Grading the quality of PTD files, and (3) PTD to Address mapping. All other steps support these three.

We care more about your logic than your tooling. Every decision must be explained.

3. Scope

- **Payer:** Centene
- **Market:** Commercial group, California
- **Network:** PPO (you must identify which plans and files make it up)
- **Provider type:** Type 1 (individual) and Type 2 (organization) NPIs

4. Data Provided

Dataset	Description	Source
Centene index files	All of Centene's index files (Ambetter, Health Net, QualChoice, Fidelis), as published	https://www.centene.com/price-transparency-files.html
NPPES extract	California NPIs with NPI type, taxonomy codes, and primary and secondary practice addresses. About 80% of addresses are stale	Refer: NPPES full download hyperlink on the URL, (Use VPN if need be) https://npiregistry.cms.hhs.gov/search
NUCC taxonomy	Maps taxonomy codes to specialties	Public
Specialty to Billing Code Mapping	Supplemented with the Case P.S. as an excel	HiLabs Supplemented

You may use or suggest other public sources, for example payer provider directory APIs, the CMS Doctors and Clinicians file, or address standardization services. Explain what each one adds.

5. Challenges

The challenges run in order, and each builds on the previous one. They are grouped under the three core problems, with supporting steps marked.

Core Problem 1: Network Mapping

#	Challenge	What to build
C1	Parse the index files	Parse Centene's index files into one table: brand, plan name, plan ID, market type and rate file URL. Many plans point to the same file, so remove duplicates.
C2	Find the PPO network	Identify the PPO plans and rate files that make up the PPO network. Plan names are inconsistent and many do not state a network type. Explain your logic and how you ruled out HMO, EPO and POS.

Core Problem 2: Grading PTD Files

#	Challenge	What to build
C3	Grade the files	Define metrics for the quality and completeness of a PTD file (e.g. code coverage, provider coverage, $0 and outlier rates, rate types, duplicates, staleness). Score every file you use and justify each metric.

Supporting Step

#	Challenge	What to build
C4	Extract rates	For the PPO network, extract rates for every provider. Where a provider has several rates for one code, state which one you keep and why.

Core Problem 3: PTD to Address Mapping

#	Challenge	What to build
C5	Map PTD TINs and NPIs to their practicing address	NPPES addresses are highly stale, so do not rely on them alone.

Supporting Step

#	Challenge	What to build
C6	Build the provider rate file	Produce one file at the NPI, TIN, billing code and address level.

Brownie Point (optional)

#	Challenge	What to build
C7	Rate cost effectiveness	Define metrics for judging whether a provider is cost effective compared with similar providers. Define "similar" and defend each metric.

Constraint: One TIN can have multiple addresses, and one provider can work at multiple addresses. The relationship from TIN to NPI to address matters. Find a way to link them, show your logic, and use or suggest additional data sources.

Using HiLabs products: You may use existing HiLabs products (e.g. MCheck, R3) as building blocks. Known approaches, such as running MCheck R3 on NPPES addresses, are a baseline, not a solution. We are looking for thinking beyond that.

6. Deliverables

#	Deliverable	Details
1	Rate file (Excel)	One row per NPI, TIN, address and billing code, with columns NPI, TIN, Address, Billing Code, Specialty and Rate
2	Logic file	Your logic for (a) identifying the PPO network and (b) mapping TIN to NPI to address. Include the sources used, the rules applied and how you handled conflicts.
3	File quality metrics	The metrics used to grade PTD